

Air Quality Arena Overview

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AQA-Data: Multi-Region Dataset

Dataset Sources

EPA (US)

CPCB (India)

AURN (UK)

CNEMC (China)

EEA_FR (France)

EEA_DE (Germany)

SINAICA (Mexico)

Dataset Statistics

Stations

3,635

Pollutants

6

Time Series

14,139

Data Processing

Cleaning

Harmonization

Missing Value Handling

2

AQA-Bench Framework

Data Split

Training (1 Year)

Testing (2 Years)

Evaluation Settings

Zero-Shot
Evaluate directly

Fine-Tuning
Adapt on 1-year data

Models Compared

Time Series Foundation Models

- VisionTS++
- TiRex
- TimesFM-2.5
- Moirai-2
- Chronos-2
- Chronos-Bolt
- Sundial
- TimesFM-2.0
- Moirai-1
- TimesFM-1.0
- Kairos

ML/DL Models

- PatchTST
- DLinear
- DeepAR
- LightGBM

Statistical Baseline Models

- AutoETS
- Seasonal Naive

3

Performance Evaluation

Evaluation Metrics

MASE

CRPS

MAE

RMSE

sMAPE

Comprehensive Comparisons

Overall Ranking

Pollutant-wise Comparison

Region-wise Comparison

Station-wise Comparison

Outputs

Leaderboard
Model Ranking

Performance Curves
Error Trends

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Error Analysis & Insights

Bias Analysis

Pollutant Bias

Country Bias

Station Bias

Urban vs. Rural Bias

Error Analysis

Geographic (Map)

Seasonal

Forecast Horizon

Error Distribution

Key Insights

- Most difficult pollutants
- Most challenging regions/stations
- TSFM strengths
- TSFM weaknesses
- Guidance for future model development